System Administration and Maintenance

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3. Hardware Components of Computer Networks

Physical Network Devices and Network Configuration

- Understand network hardware components.
- Explain the function of network devices.
- Configure basic network settings.
- Understand hosts, routers, and network configuration servers.

Learning Outcomes

* At the end of this lecture students will be able to Demonstrate an understanding of the components of a Computer Network:

  - Identify computer network hardware
  - Explain the role of networking devices
  - Describe how hosts communicate
  - Configure network parameters
  - Explain routers and switches
  - Understand DHCP and DNS servers
  - Perform basic network troubleshooting

What Is Computer Network Hardware

* Network hardware consists of all physical devices used to create and maintain a computer network.

* **It includes:**
  - Computers (Hosts)
  - Network Interface Cards (NIC)
  - Hubs
  - Switches
  - Routers
  - Wireless Access Points
  - Modems
  - Servers
  - Firewalls
  - Network Cables

Computers (Hosts)



Network Interface Cards (NIC)



Hubs



Switches



Routers



Wireless Access Points



Modems



Servers



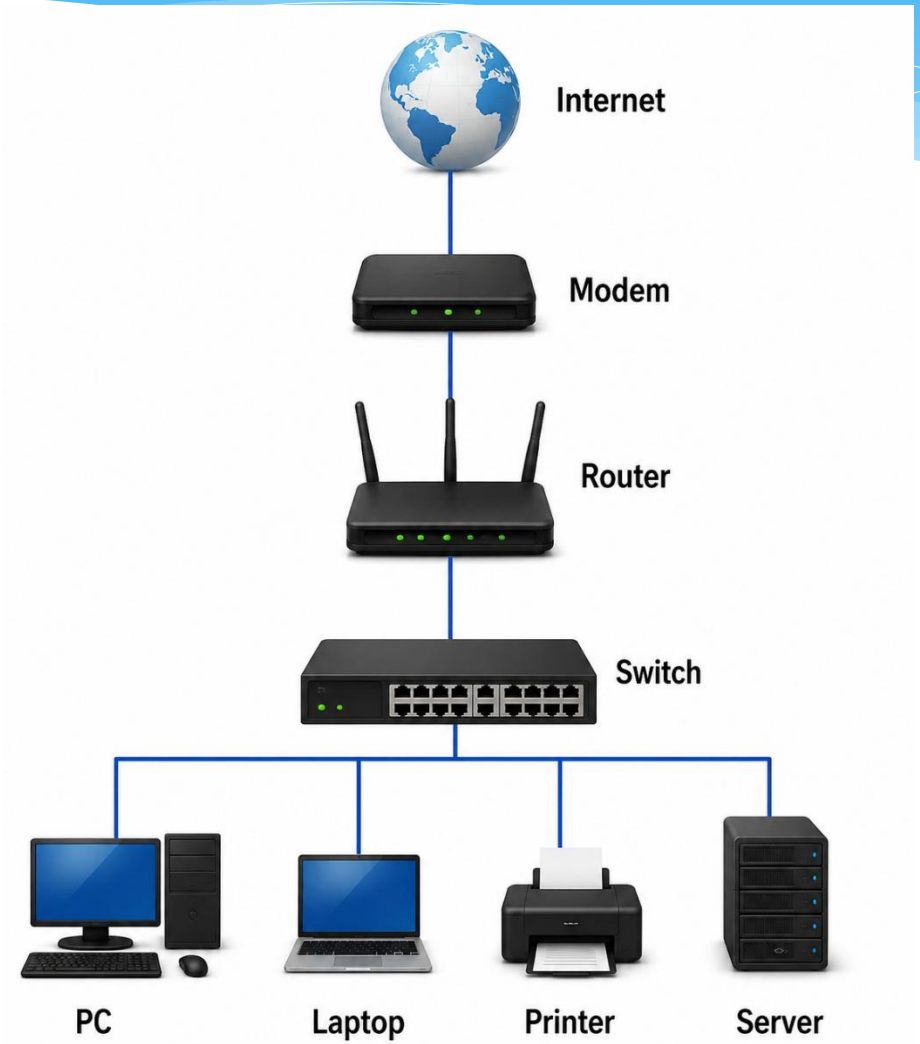
Firewalls



Network Cables



Computer Network Overview



Main Hardware Components

DEVICE	PURPOSE
Host	End user computer
NIC	Connects device to network
Switch	Connects devices in LAN
Router	Connects different networks
Modem	Connects ISP (Internet Service Provider)
Wireless AP	Wi-Fi connectivity
Firewall	Security
Server	Provides services

Computer/Hosts/Node/Client

* A **Host** is any device connected to the network.

  Eg:
  - Desktop
  - Laptop
  - Server
  - Smartphone
  - Printer
  - CCTV Camera
  - IoT Device

* Each host has
  - IP Address (Logical Address) eg: 192.168.1.1
  - MAC Address (Physical Address)
  - Host Name

Network Interface Card (NIC)

NIC enables a computer to communicate with the network.

* Types
  - Ethernet NIC
  - Wireless NIC

* Functions
  - Sends data
  - Receives data
  - Stores MAC Address

Eg:

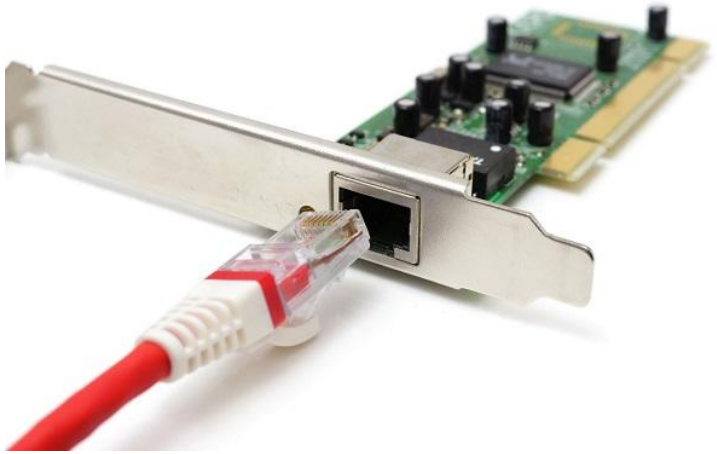
Computer

|

Ethernet Cable

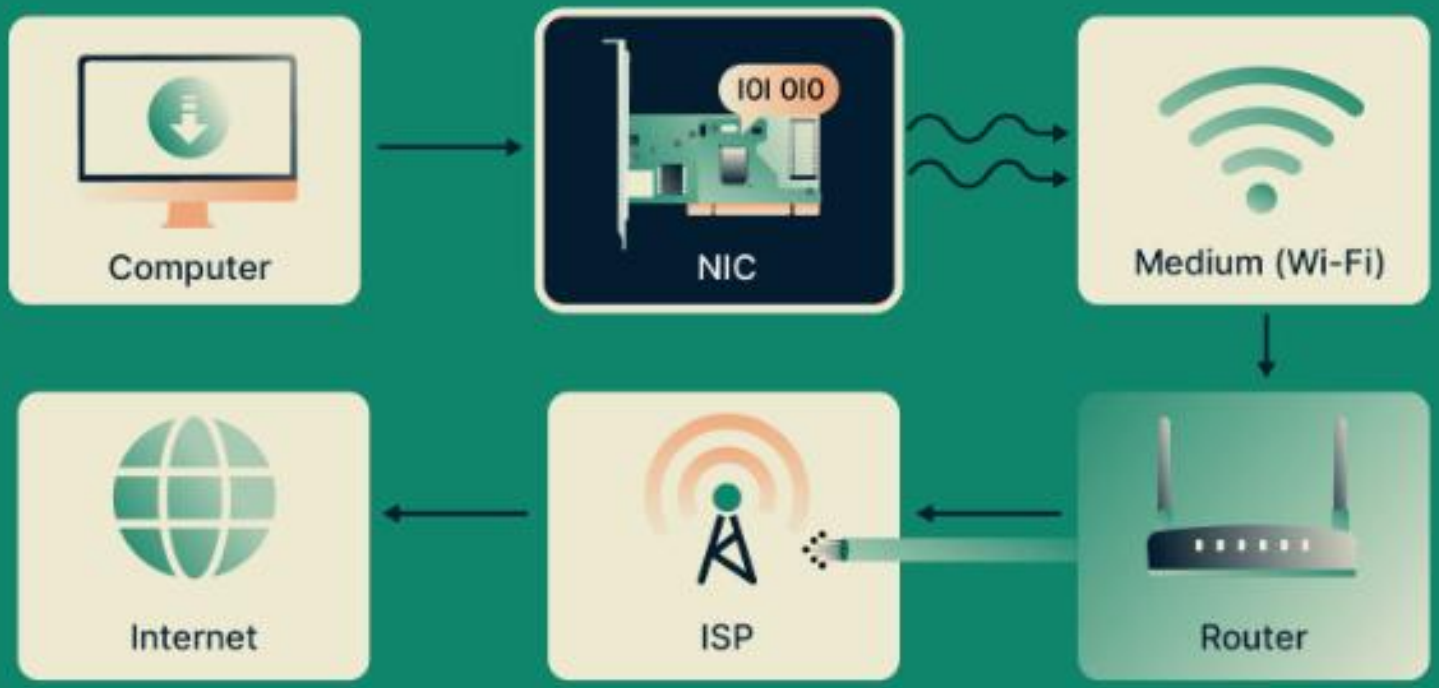
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Switch

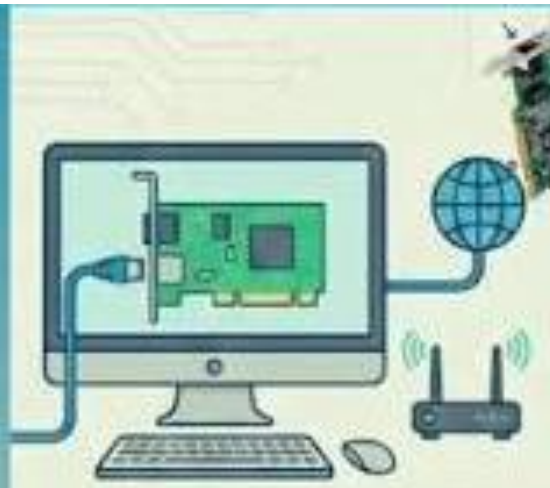


How a NIC connects a device to the internet

The NIC acts as the bridge between a computer and the router, enabling communication with the ISP and access to the internet



The NIC converts digital data into network signals, which a router forwards to the ISP and wider internet



NIC = Network Interface Card

A hardware device that connects a computer to a network.

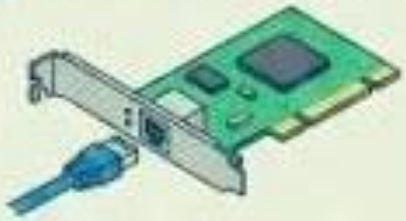
It allows your computer to send and receive data over a network.

Why is NIC Important?

- Connect to the **Internet**
- Join a **Local Area Network (LAN)**
- Communicate with a **Switch or Router**

Types of NIC

1 Ethernet NIC (Wired)



Uses LAN cable (RJ45)

2 Wireless NIC (Wi-Fi)



Connects without cable

MAC Address

* Every NIC has a unique MAC address.

  Eg: 00:1A:2B:3C:4D:5E

* **Used for**
  - Device identification
  - Local communication
  - Network management

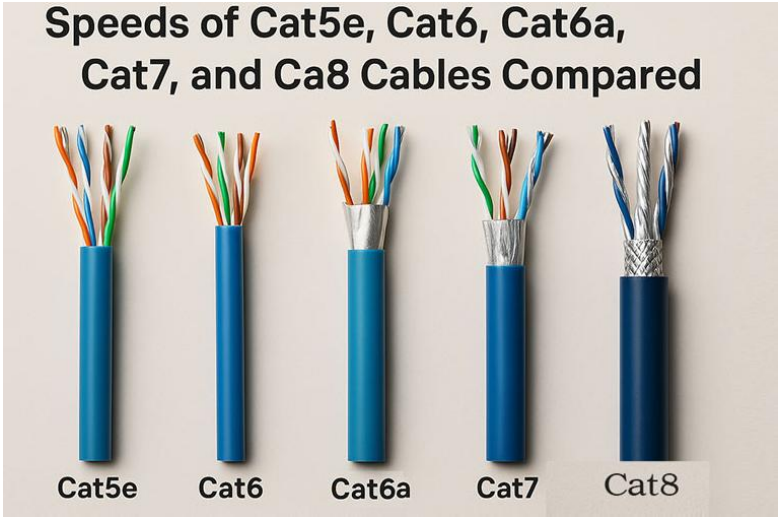
Ethernet Cables

Most common network media.

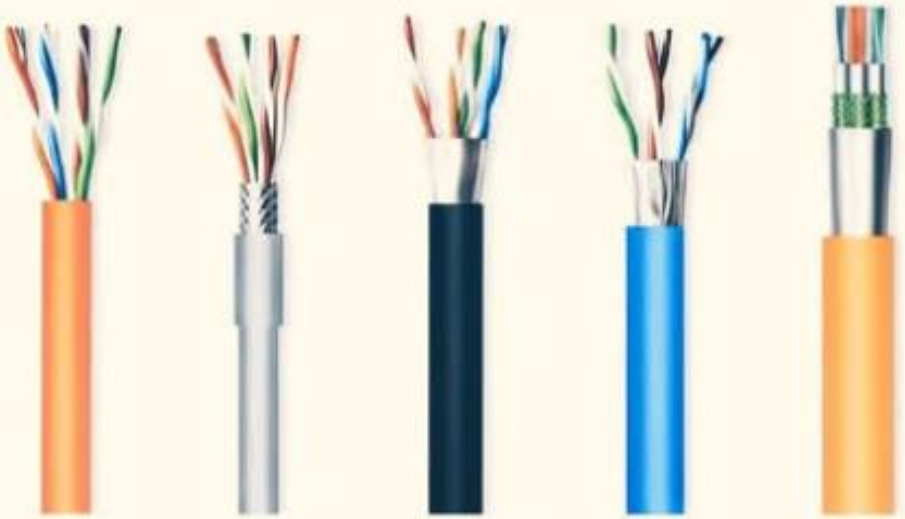


* **Types**
  - Cat5e
  - Cat6
  - Cat6a
  - Cat7
  - Fiber Optic

* **Advantages**
  - High speed
  - Reliable
  - Low interference



Speeds of Cat5e, Cat6, Cat6a,Cat7, and Cat8 LAN Cables Compared



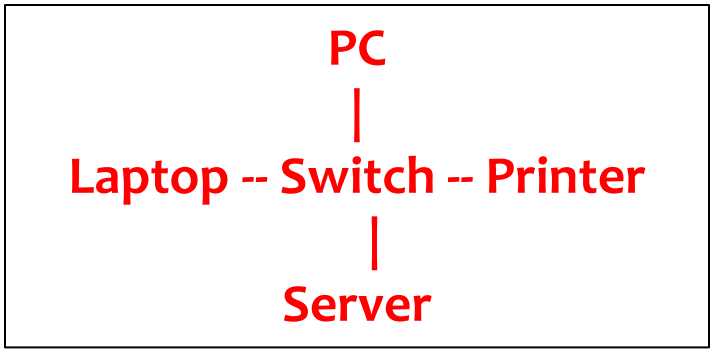


Max Data Transfer Speed	Max Bandwidth	Optimal Cable Length for Max Speed	Common Use Case
1 Gbps	100 MHz	Up to 100 meters	Residential networking, basic office use
10 Gbps	250 MHz	Up to 55 meters	General office networking, some data center applications
10 Gbps	500 MHz	Up to 100 meters	Advanced office netw-orta, data centers, industrial applications
10 Gbps	600 MHz	Up to 100 meters	High-speed networks, data centers server rooms
25-40 Gbps	2000 MHz	Up to 30 meters	High-performance data centers server-to-server conn

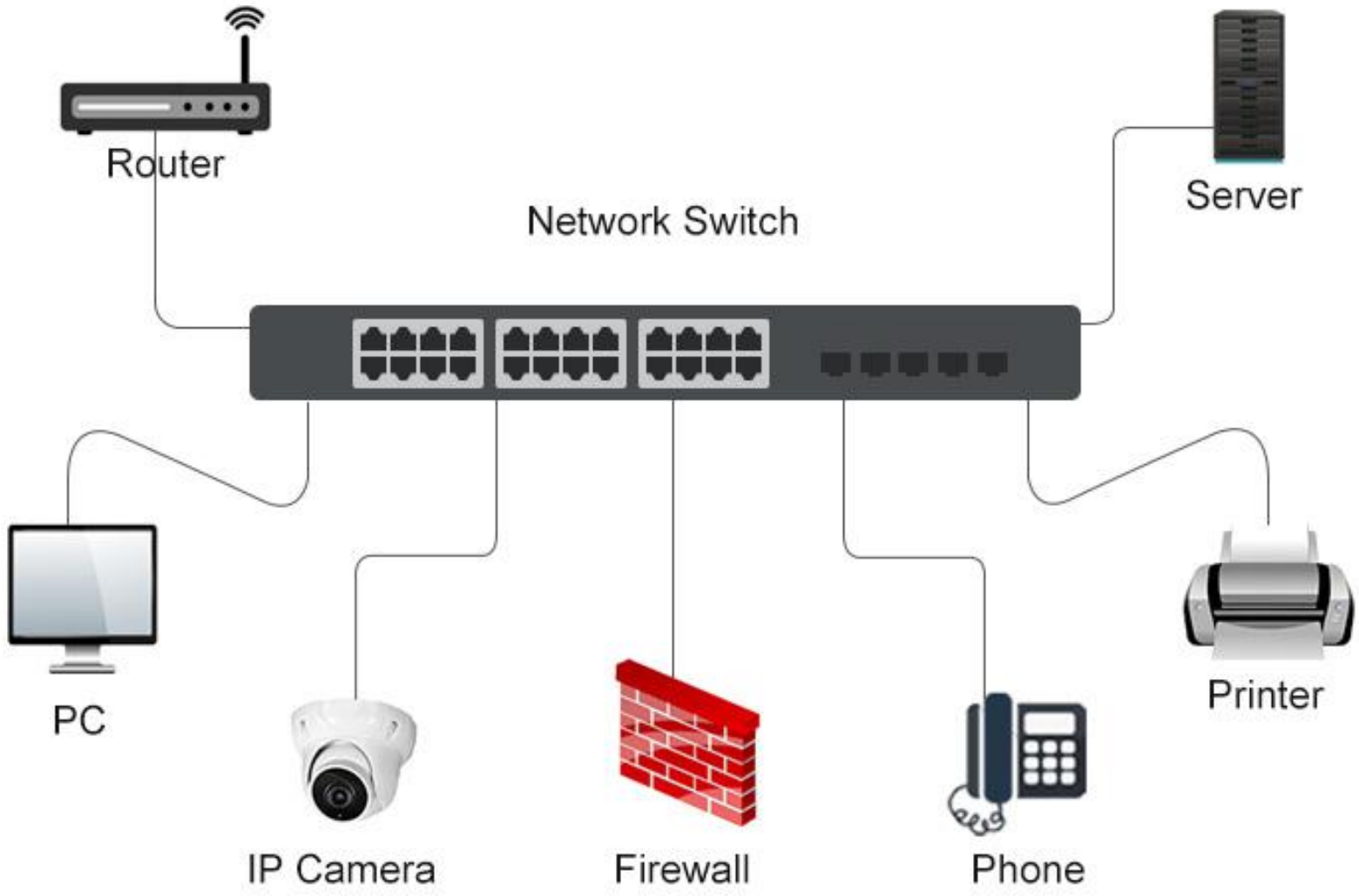
Switch

A switch connects multiple devices in a LAN.

* Functions
  - Learns MAC addresses
  - Sends data only to destination device
  - Improves performance






Router
Server
Network Switch
PC
IP Camera
Firewall
Phone
Printer

Hub vs Switch

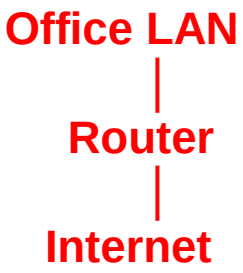
Hub	**Switch**
Broadcasts all data	Sends only to destination
Slow	Fast
Shared bandwidth	Dedicated bandwidth
Old technology	Modern technology

Hub  VS Switch 

A hub broadcasts all incoming data to every connected device.	A switch learns device MAC addresses and forwards data only to the intended port.
A hub operates at the Physical Layer handling raw electrical signals without interpreting data.	A switch operates at the Data Link Layer understanding MAC addresses and processing data in frames.
All devices connected to a hub share a single collision domain.	Each port on a switch creates its own separate collision domain.
A hub lacks intelligence, leading to inefficient bandwidth and slower performance.	A switch is intelligent, learning MAC addresses to efficiently manage data flow.

Router

* Router connects different networks.
* Eg:



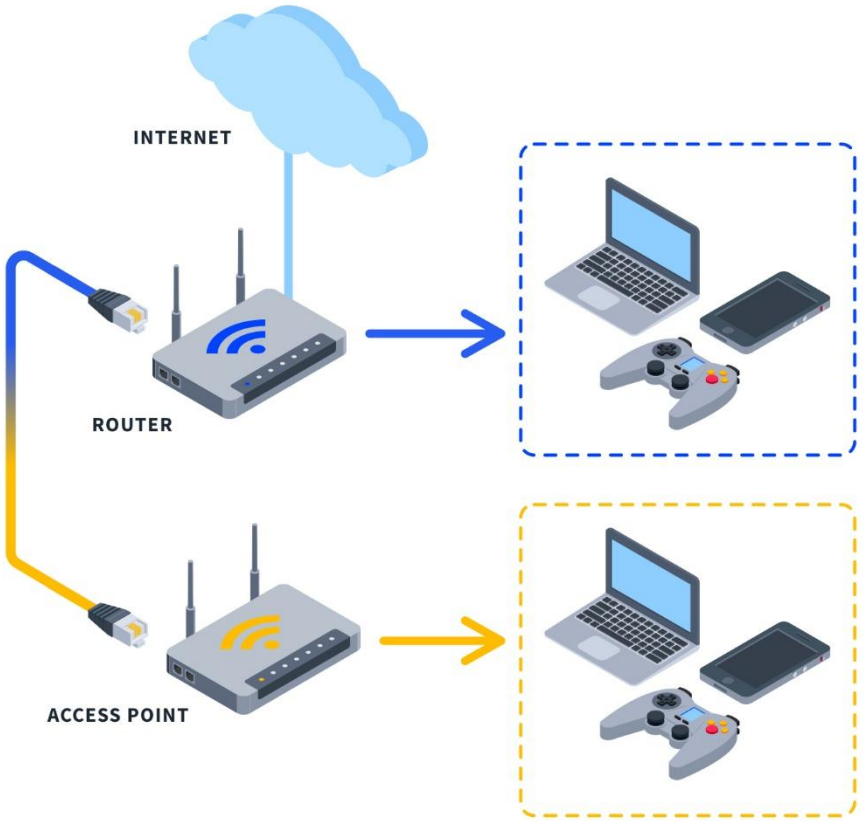
* Functions
  - Packet forwarding
  - Routing
  - NAT
  - Internet sharing
  - DHCP


1PPS INPUT
1PPS OUTPUT
10MHz INPUT
10MHz OUTPUT
ALM
SYS
MST
ID
RESET

Wireless Access Point

Provides wireless connectivity.

* Devices connected
  - Mobile phones
  - Tablets
  - Laptops
  - Smart TVs
* Standards
  - Wi-Fi 5
  - Wi-Fi 6
  - Wi-Fi 6E
  - Wi-Fi 7

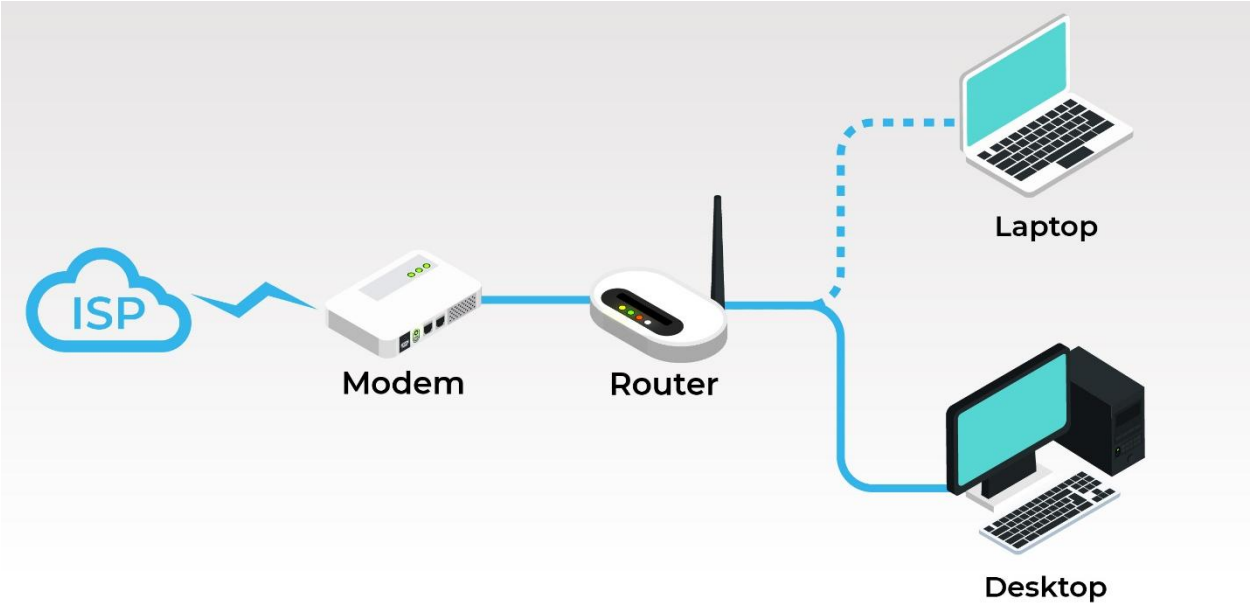


Modem

Modem connects your network to the ISP.

* Types
  - DSL
  - Cable
  - Fiber
  - Cellular

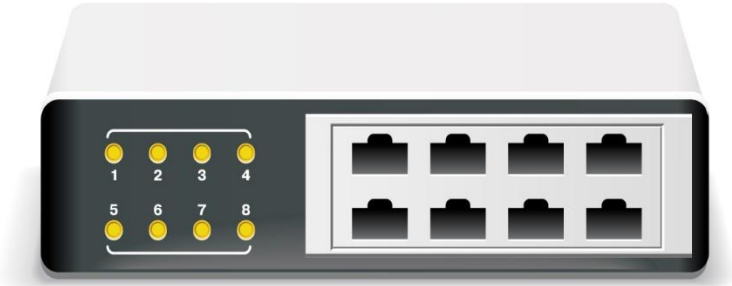
ISP
|
Modem
|
Router
|
LAN



Modem vs Router vs Switch







Network Servers

A server provides services to clients.

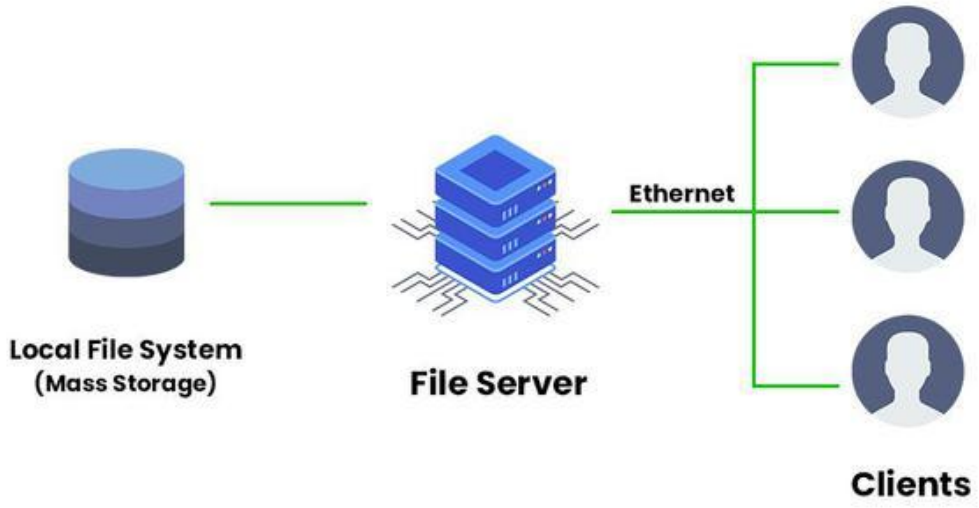
* Eg:
  - File Server
  - Print Server
  - Mail Server
  - Web Server
  - Database Server
  - DNS Server
  - DHCP Server



File Server

Stores shared files

* Advantages
  - Central storage
  - Easy backup
  - Access control
  - Data sharing

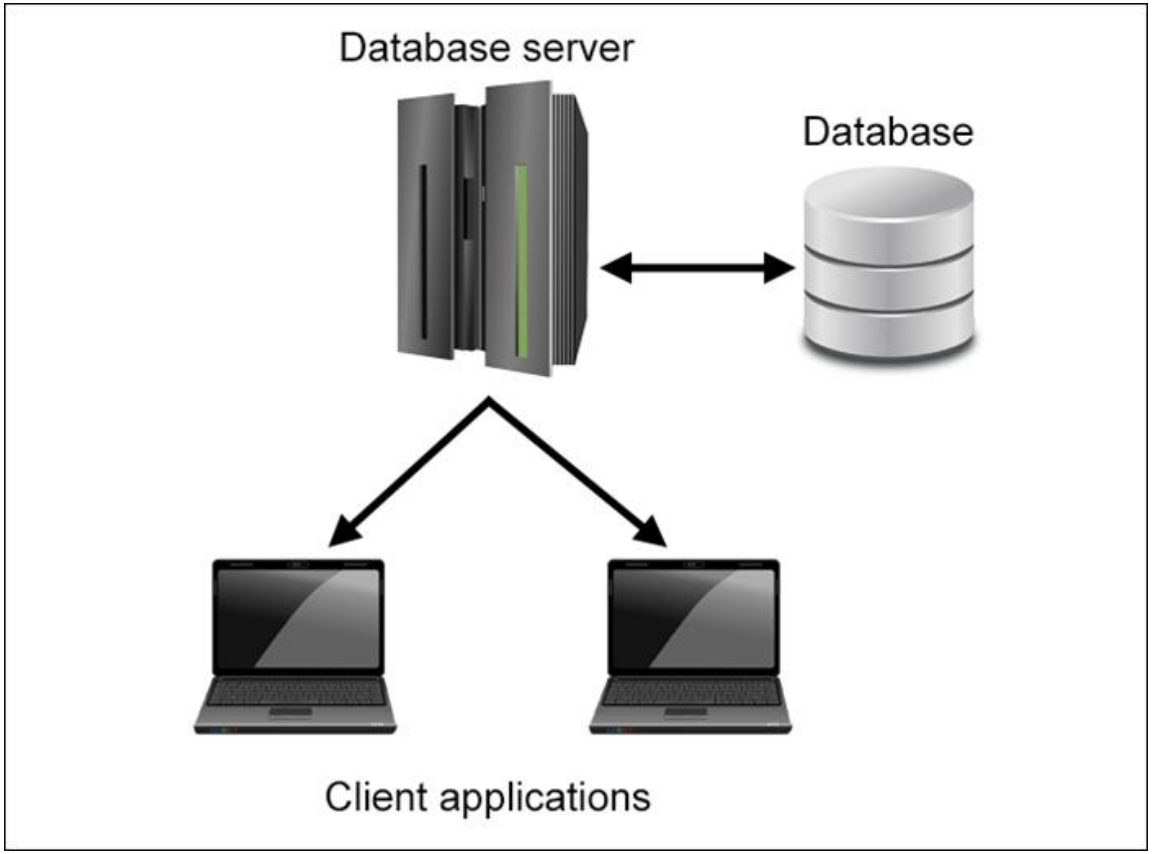


Database Server

Stores organizational data

* Eg:
  - Student records
  - Hospital systems
  - Banking systems
  - ERP systems

* Popular databases
  - MySQL
  - PostgreSQL
  - Microsoft SQL Server
  - Oracle



DHCP Server

DHCP(Dynamic Host Configuration Protocol)

* Automatically assigns
  - IP Address
  - Subnet Mask
  - Gateway
  - DNS Server

* Eg:

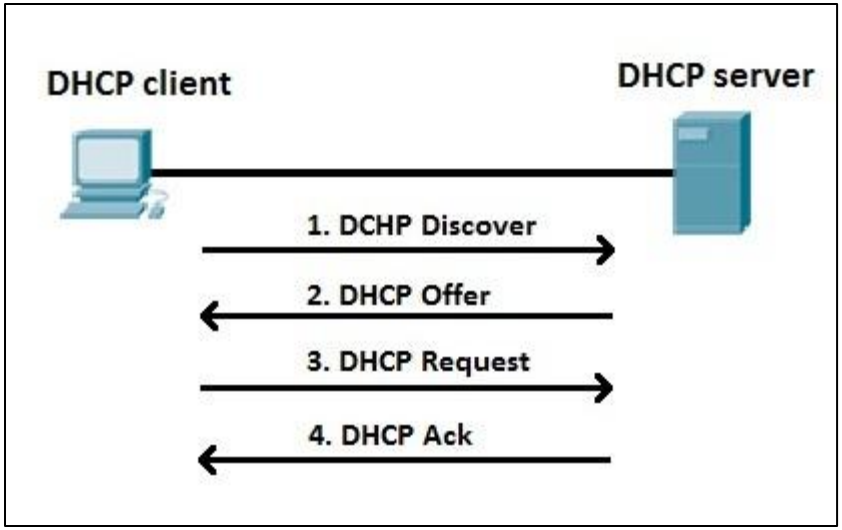
Client

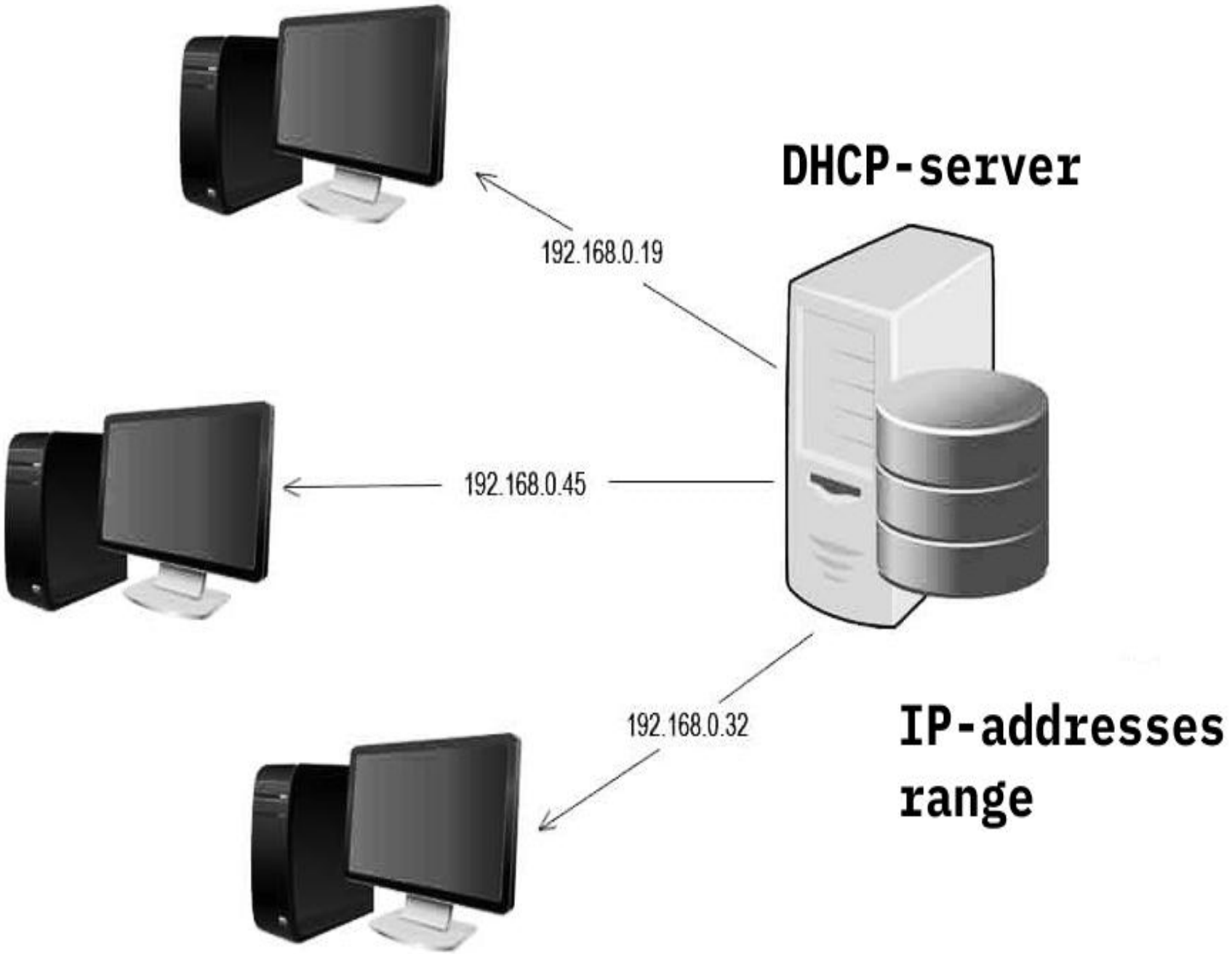
| Request

DHCP Server

| Assign IP

192.168.1.100




DHCP-server
192.168.0.19
192.168.0.45
192.168.0.32
IP-addresses
range

DNS Server

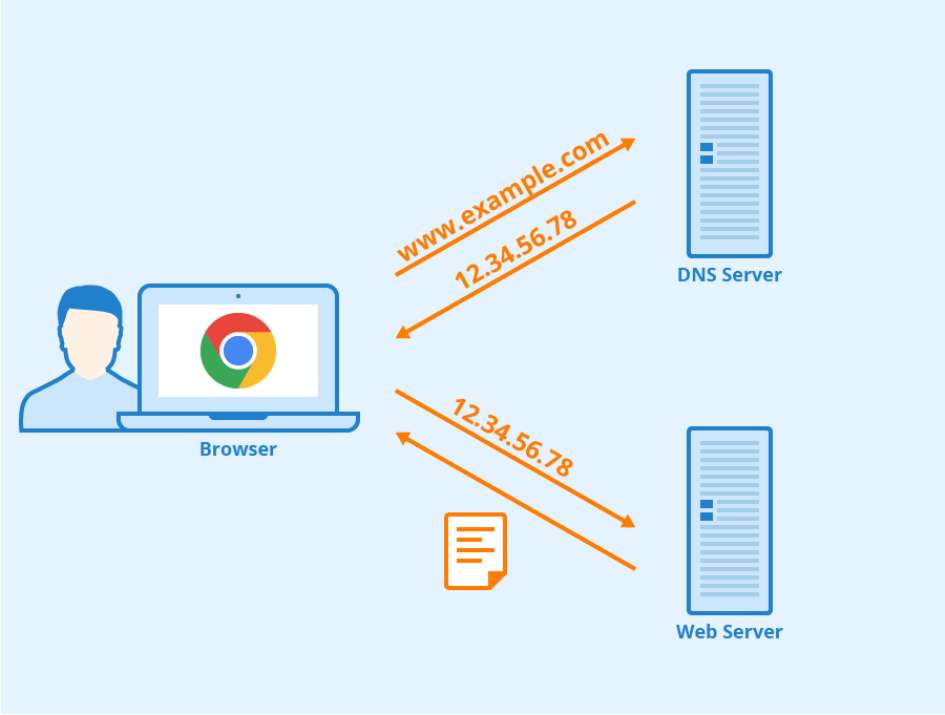
DNS converts domain names into IP addresses.

* Eg:

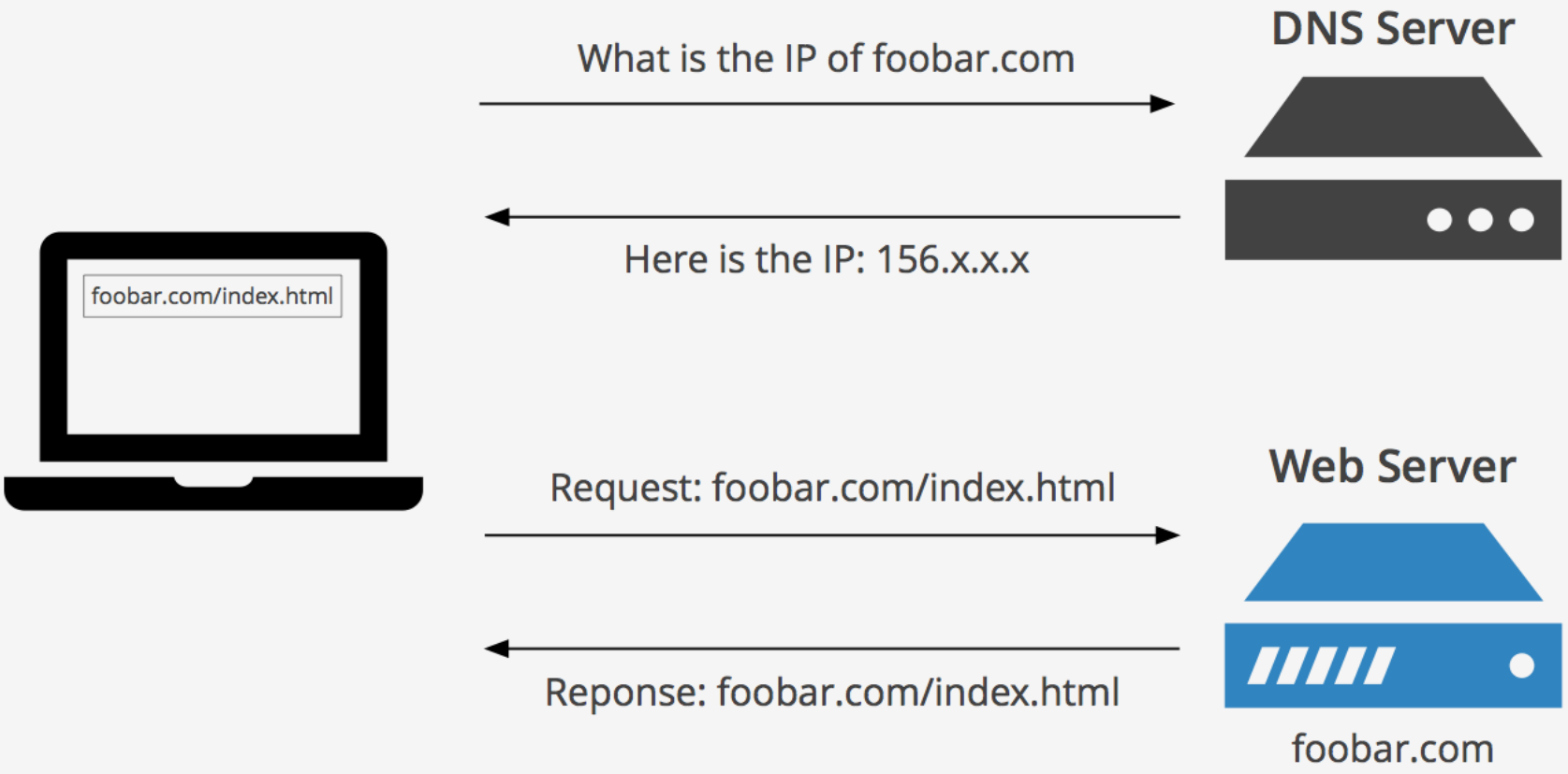
  www.google.com

  ↓

  142.250.xxx.xxx



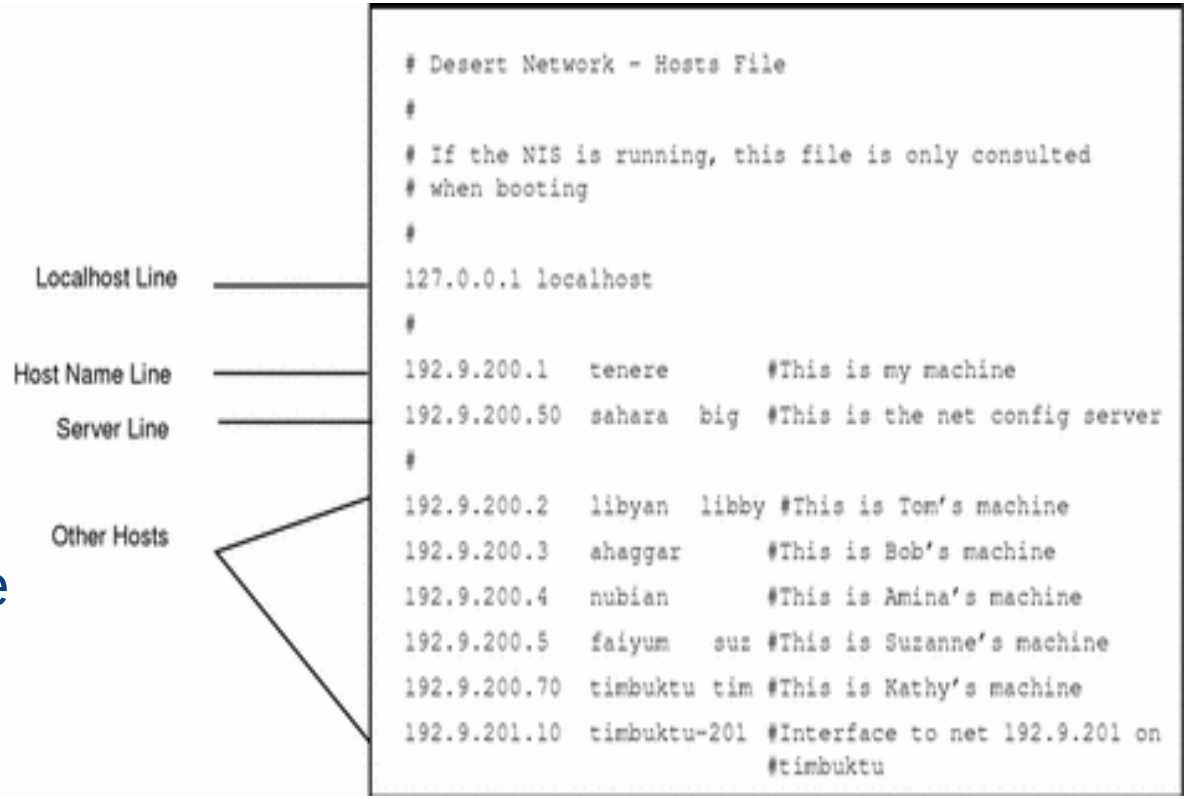
* Without DNS, users must remember IP addresses.


DNS Server
What is the IP of foobar.com
Here is the IP: 156.x.x.x
foobar.com/index.html
Web Server
Request: foobar.com/index.html
Reponse: foobar.com/index.html
foobar.com

Network Configuration Files

Network configuration defines how a device communicates.

* Includes
  - IP Address
  - Gateway
  - DNS
  - Hostname
  - Routing
* Configuration can be
  - Static
  - Dynamic (DHCP)



Static vs Dynamic Configuration

Static	**Dynamic**
Manual	Automatic
Fixed IP	DHCP
More control	Easy management
Suitable for servers	Suitable for clients

Hosts File

Hosts file maps hostnames to IP addresses.

* Example

192.168.1.20 webserver

192.168.1.21 database

Used before DNS lookup.

Router Configuration

* Basic router settings include
  - Hostname
  - Interfaces
  - IP Address
  - Routing Table
  - NAT
  - DHCP
  - Firewall Rules
* Example

```
Router(config)# hostname KDU-R1
```

Network Configuration Servers

Common servers

* **DHCP Server**
  - Assigns IP
* **DNS Server**
  - Name resolution
* **RADIUS Server (Remote Authentication Dial-In User Service)**
  - Authentication
* **NTP (Network Time Protocol) Server**
  - Time synchronization
* **Active Directory (AD)**
  - User authentication

RADIUS Server (Remote Authentication Dial-In User Service)

- A centralized **AAA (Authentication, Authorization and Accounting)** protocol
- Used to control access to network resources
- Commonly used in:
  - Enterprise Wi-Fi
  - VPNs
  - Network access control
  - Remote access services
- Uses a **client–server architecture**

What is RADIUS

* **RADIUS** provides centralized management of network user access.
* **Main functions:**
  1. **Authentication** – Who is the user?
  2. **Authorization** – What is the user allowed to access?
  3. **Accounting** – What network resources did the user use?

Eg: **Student → Wi-Fi Access Point → RADIUS Server → User Database**

How RADIUS Works

* **Basic authentication process:**

1. User connects to the network.
2. Network Access Server (NAS) requests user credentials.
3. NAS sends an **Access-Request** to the RADIUS server.
4. RADIUS verifies the credentials.
5. Server responds:
   - **Access-Accept** → User is allowed
   - **Access-Reject** → Access is denied
   - **Access-Challenge** → Additional authentication required

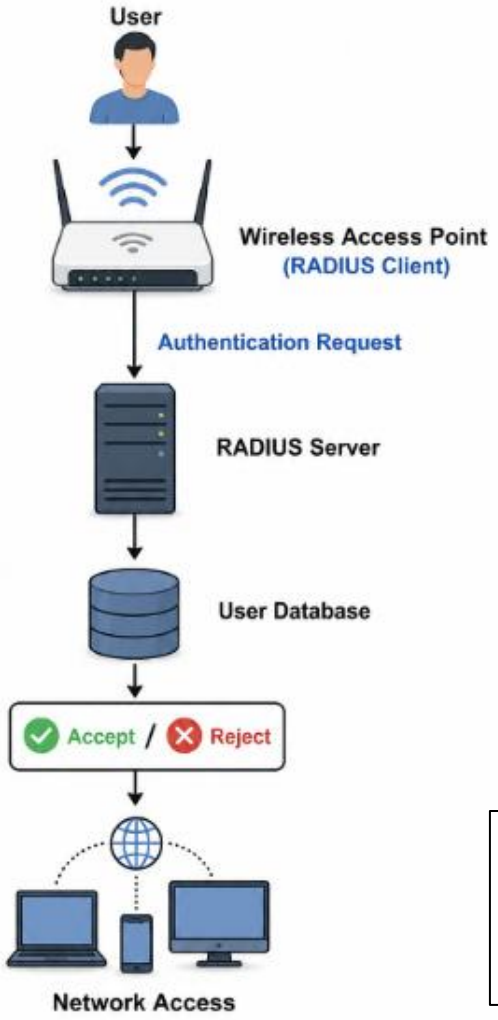
RADIUS Components

Component	Function
• **RADIUS Client / NAS**	Requests authentication
• **RADIUS Server**	Authenticates and authorizes users
• **User Database**	Stores user credentials and policies
• **Network User**	Requests access to the network

Example for RADIUS clients:

- ✓ Wireless Access Point
- ✓ VPN Gateway
- ✓ Router
- ✓ Network Access Server

RADIUS Authentication Example Enterprise Wi-Fi scenario

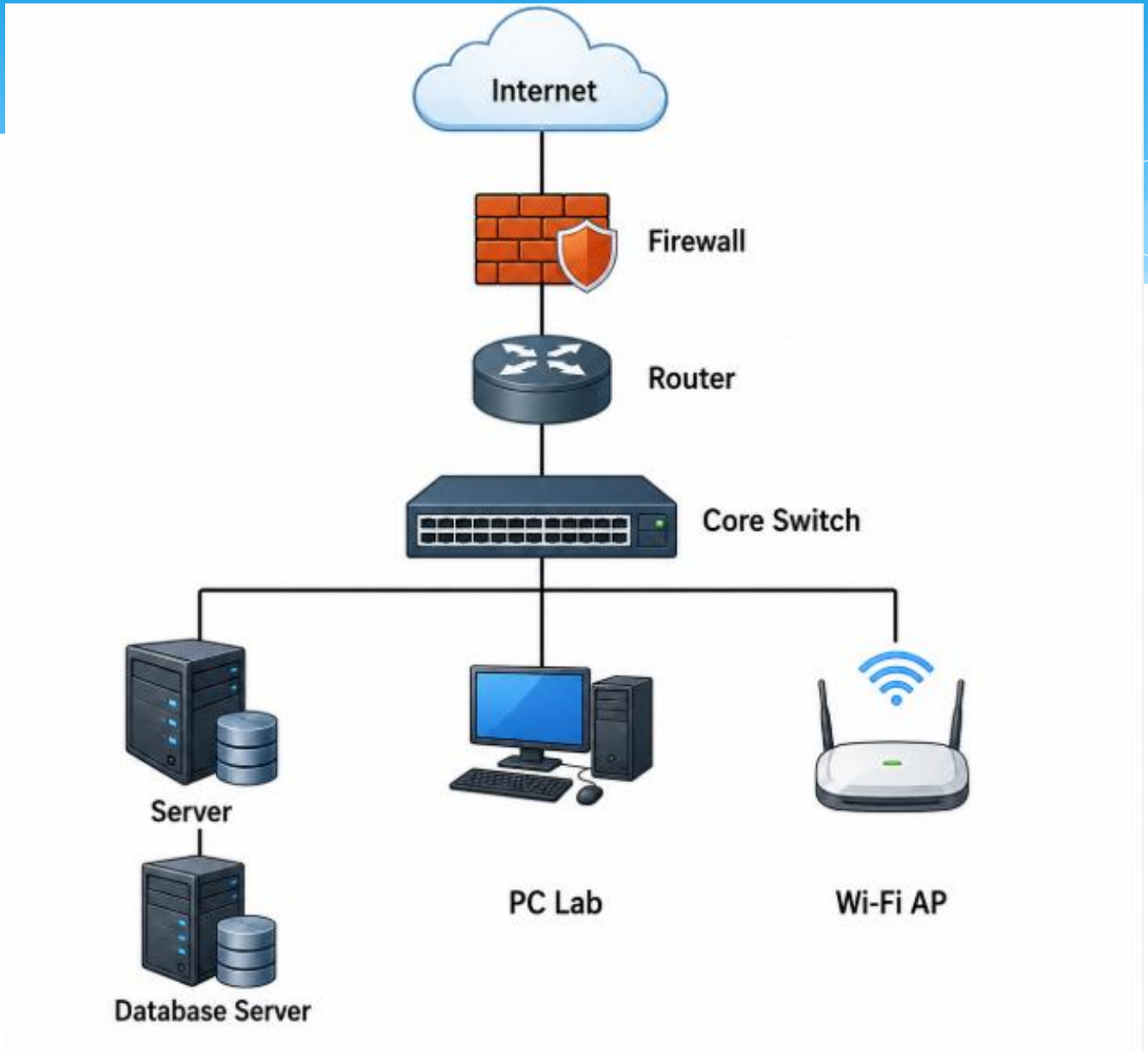


- A university can use RADIUS to authenticate students and staff before allowing access to campus Wi-Fi.

Advantages of RADIUS

- **Centralized authentication**
- Improved **network security**
- Centralized user and access policies
- Supports **AAA** (Authentication, Authorization, and Accounting)
- Suitable for large organizations
- Easier user management
- Can integrate with **LDAP** (Lightweight Directory Access Protocol) **, Active Directory, and other identity systems**

Network Topology Example



Common Network Problems

Problem	Cause
No Internet	Router failure
Duplicate IP	Static IP conflict
Slow Network	Bad switch
No Wi-Fi	Access Point issue
Cannot open website	DNS failure

Practical Laboratory Exercise

* Students should perform:

- Task 1
  - View IP configuration
    - Windows
      - ✓ **Ipconfig**
    - Linux
      - ✓ **ifconfig**
        or
      - ✓ **ip addr**

- Task 2
  - Test connectivity
    - ✓ **ping google.com**

- Task 3
  - Trace network
    - ✓ **tracert google.com**

Classroom Activity (20 Minute)

1. Draw a network topology for a university computer laboratory.

1. Identify all required hardware components.

2. Assign IP addresses to hosts.

3. Explain the role of the router, switch, DHCP server, and DNS server.

4. Present the design to the class.

Questions

1. What is the difference between a host and a server?
2. Why is a switch preferred over a hub?
3. What happens if the DHCP server fails?
4. Why is DNS important for users?
5. What are the advantages of static IP addressing for servers?
6. How does a router differ from a switch?
7. What information is typically stored in a network configuration file?
8. What network problems can result from incorrect IP configuration?

1. What is the primary function of a Network Interface Card (NIC)?
2. Which device connects different networks together?
3. What does DHCP stand for, and what is its purpose?
4. What is the role of a DNS server?
5. Give two examples of network configuration servers.
6. State one advantage and one disadvantage of static IP addressing.
7. Which command can be used to display IP configuration in Windows?
8. Explain the purpose of the hosts file in a computer system.

THANK YOU